Implementation-of-Erosion-and-Dilation

DEVELOPED BY :M.VIRUMAA HARISH

REG NO : 212223230246

Aim

To implement Erosion and Dilation using Python and OpenCV.

Software Required

1. Anaconda - Python 3.7
2. OpenCV

Algorithm:

Step 1:

Import necessary libraries, including OpenCV (cv2) and Matplotlib (plt), to load, manipulate, and display images.

Step 2:

Use cv2.putText() to add text to the image at a specific location with chosen font, size, color, and thickness.

Step 3:

Define a structuring element using cv2.getStructuringElement() to specify the shape and size for morphological transformations.

Step 4:

Apply erosion to the image using cv2.erode() with the structuring element to shrink white regions and reduce noise.

Step 5:

Dilate the eroded image using cv2.dilate() with the structuring element to expand white regions and enhance features.

Step 6:

Display the original and processed images using plt.imshow() with proper axis configuration and titles for comparison.

Step 7:

Finalize by calling plt.show() to display all images in a single figure for easy visualization and comparison.

Program:

Import the necessary packages

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image = np.zeros((500, 500, 3), dtype="uint8")
```

Create the Text using cv2.putText

```
text = "M.Suren."
font = cv2.FONT_HERSHEY_SIMPLEX
cv2.putText(image, text, (50, 150), font, 2, (255, 255, 255), 3)
```

Create the structuring element

```
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))
```

Original image

```
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
```

Erode the image

```
eroded_image = cv2.erode(image, kernel, iterations=1)
eroded_image_rgb = cv2.cvtColor(eroded_image, cv2.COLOR_BGR2RGB)
plt.imshow(eroded_image_rgb)
plt.title("Eroded Image")
plt.axis("off")
```

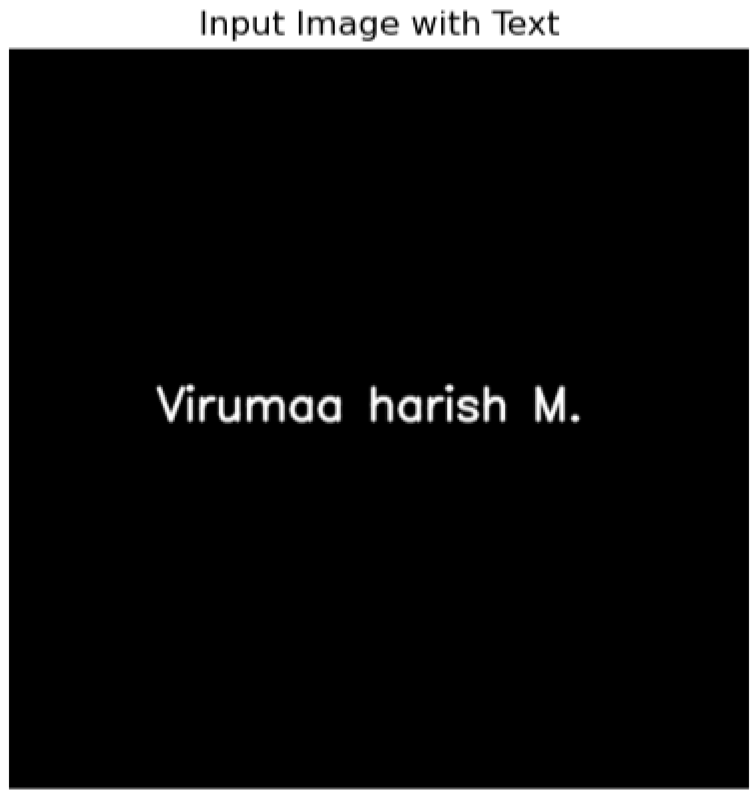
Dilate the image

```
dilated_image = cv2.dilate(image, kernel, iterations=1)
dilated_image_rgb = cv2.cvtColor(dilated_image, cv2.COLOR_BGR2RGB)
```

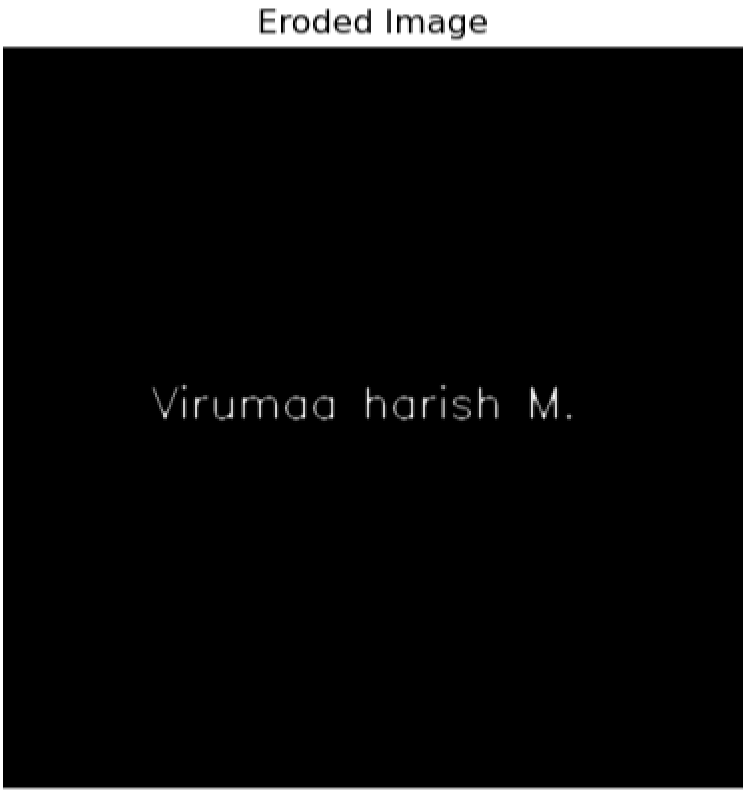
```
plt.imshow(dilated_image_rgb)
plt.title("Dilated Image")
plt.axis("off")
```

Output:

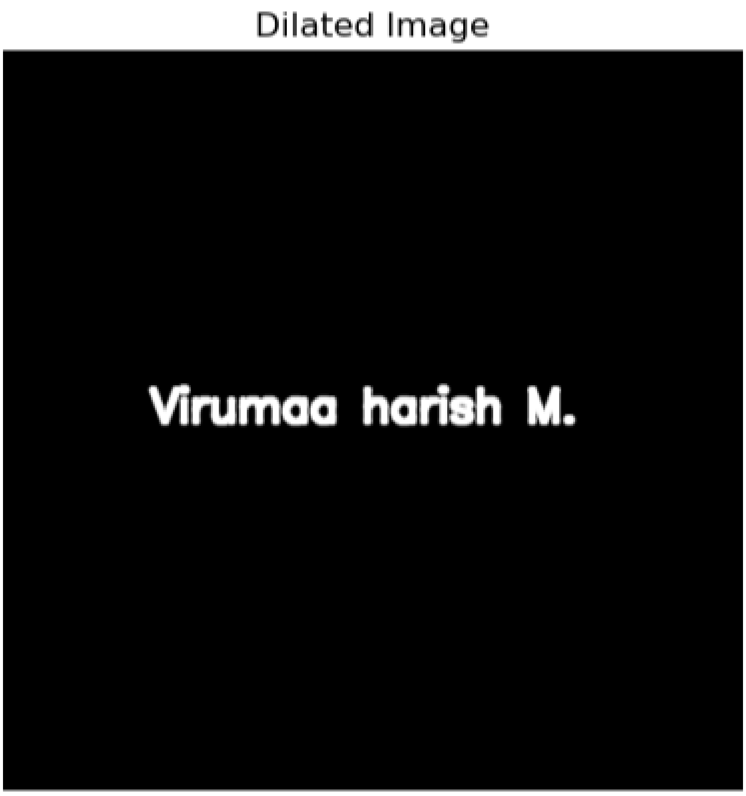
Display the input Image



Display the Eroded Image



Display the Dilated Image



Result

Thus the generated text image is eroded and dilated using python and OpenCV.